

AWS & Active Directory Lab Report

Part 1: AWS Infrastructure Setup

Step 1: Create VPC

Create a custom VPC (e.g. 10.0.0.0/24).

Step 2: Create Internet Gateway

Create an IGW and attach it to the VPC.

Step 3: Create Public Subnet

Create a subnet inside the VPC.

Step 4: Enable Auto Assign Public IPv4

Enable auto-assign public IP for the subnet.

Step 5: Create Route Table

Create a route table.

Step 6: Configure Route

Add route 0.0.0.0/0 -> Internet Gateway.

Step 7: Associate Route Table

Associate the route table with the public subnet.

Step 8: Create Security Group

Allow TCP 3389 (RDP), 80, 443, and 22 if required.

Step 9: Launch Windows EC2

Choose the public subnet, assign public IP, attach the security group.

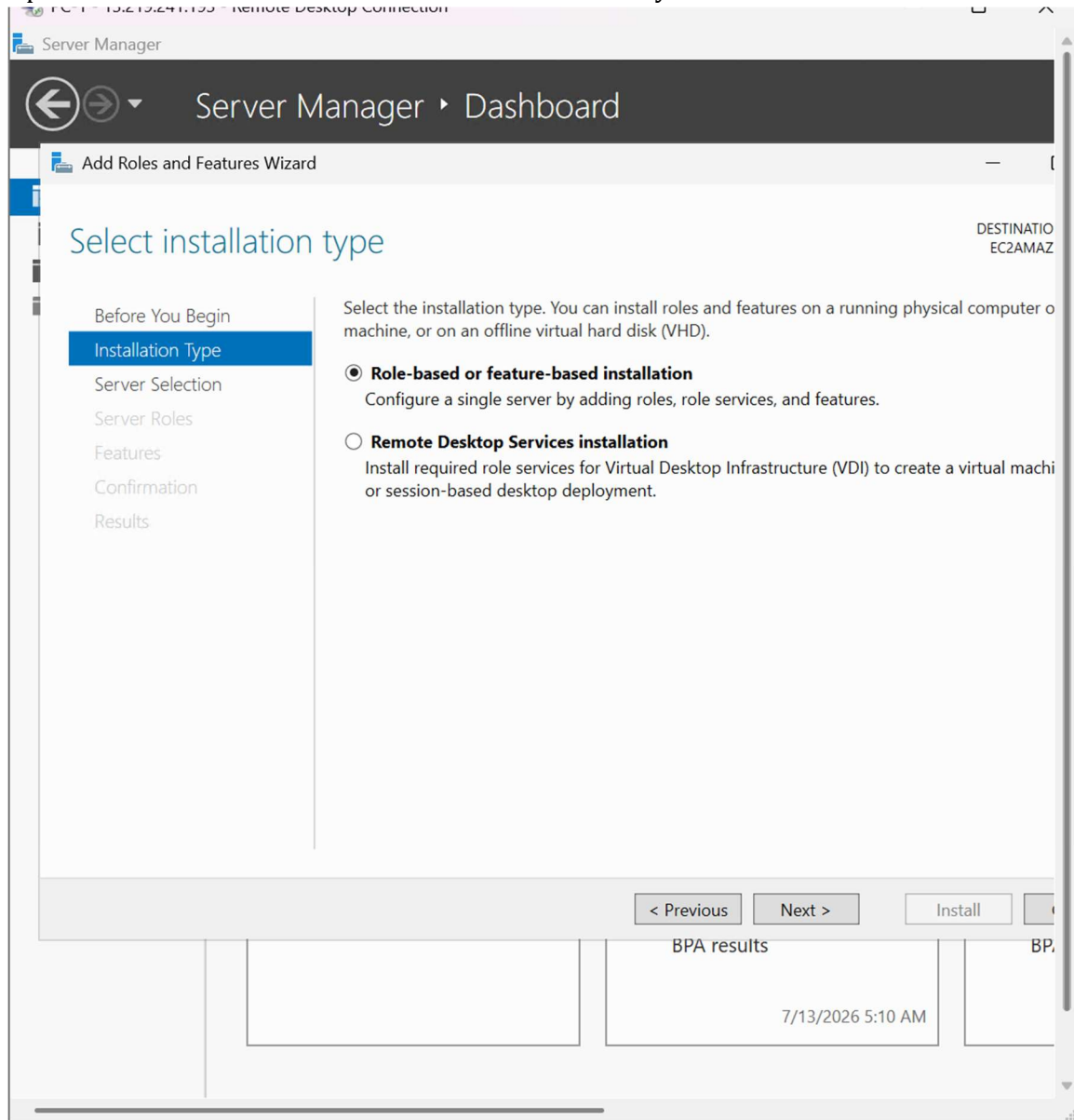
Step 10: Connect via RDP

Retrieve Administrator password and connect using Remote Desktop.

Part 2: Install Active Directory Domain Services

AD Step 1: Install AD DS Role

Open Add Roles and Features and select Active Directory Domain Services.



AD Step 2: Create New Forest

Promote the server and create a new forest named lab.local.

Server Manager

Server Manager ▶ Dashboard

Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
EC2AMAZ-NR758OI

✖ An Active Directory domain controller for the domain "lab.local" could not be contacted. [Show more](#) ✖

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

☐ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

< Previous Next > Install Cancel

All Servers 1

AD Step 3: Domain Controller Options

Keep DNS Server and Global Catalog enabled and configure the DSRM password.

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' in the 'Server Manager' application. The window title is 'Server Manager - Dashboard'. The wizard is titled 'Domain Controller Options' and is for 'TARGET SERVER EC2AMAZ-NR758OI'. The left sidebar shows the progression: Deployment Configuration, Domain Controller Options (selected), DNS Options, Additional Options, Paths, Review Options, Prerequisites Check, Installation, and Results. The main content area is titled 'Select functional level of the new forest and root domain'. It has two dropdown menus: 'Forest functional level:' and 'Domain functional level:', both set to 'Windows Server 2025'. Below this is the 'Specify domain controller capabilities' section with three checkboxes: 'Domain Name System (DNS) server' (checked), 'Global Catalog (GC)' (checked), and 'Read only domain controller (RODC)' (unchecked). The next section is 'Type the Directory Services Restore Mode (DSRM) password', with fields for 'Password:' and 'Confirm password:', both masked with dots. A link 'More about domain controller options' is at the bottom. The bottom of the wizard has buttons for '< Previous', 'Next >', 'Install', and 'Cancel'. At the very bottom, a red bar shows 'All Servers 1'.

Server Manager - Dashboard

Active Directory Domain Services Configuration Wizard

Domain Controller Options

TARGET SERVER
EC2AMAZ-NR758OI

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2025

Domain functional level: Windows Server 2025

Specify domain controller capabilities

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

[More about domain controller options](#)

< Previous Next > Install Cancel

All Servers 1

AD Step 4: NetBIOS Name

Accept the default NetBIOS name LAB.

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar indicates it is part of the 'Server Manager' and 'Dashboard'. The main heading is 'Additional Options'. On the right, it identifies the 'TARGET SERVER' as 'EC2AMAZ-NR758OI'. A sidebar on the left lists the wizard steps: 'Deployment Configuration', 'Domain Controller Options', 'DNS Options', 'Additional Options' (which is highlighted in blue), 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main content area contains the instruction 'Verify the NetBIOS name assigned to the domain and change it if necessary' and a label 'The NetBIOS domain name:' followed by a text input field containing the value 'LAB'. Below this, there is a link that says 'More about additional options'. At the bottom of the wizard, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'. At the very bottom of the window, there is a red status bar with a server icon, the text 'All Servers', and the number '1'.

Server Manager

Server Manager ▸ Dashboard

Active Directory Domain Services Configuration Wizard

Additional Options

TARGET SERVER
EC2AMAZ-NR758OI

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name:

[More about additional options](#)

< Previous Next > Install Cancel

All Servers 1

AD Step 5: Database Paths

Keep the default NTDS, Logs and SYSVOL locations.

Server Manager

Server Manager > Dashboard

Active Directory Domain Services Configuration Wizard

Paths

TARGET SERVER
EC2AMAZ-NR758OI

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Specify the location of the AD DS database, log files, and SYSVOL

Database folder: C:\Windows\NTDS ...

Log files folder: C:\Windows\NTDS ...

SYSVOL folder: C:\Windows\SYSVOL ...

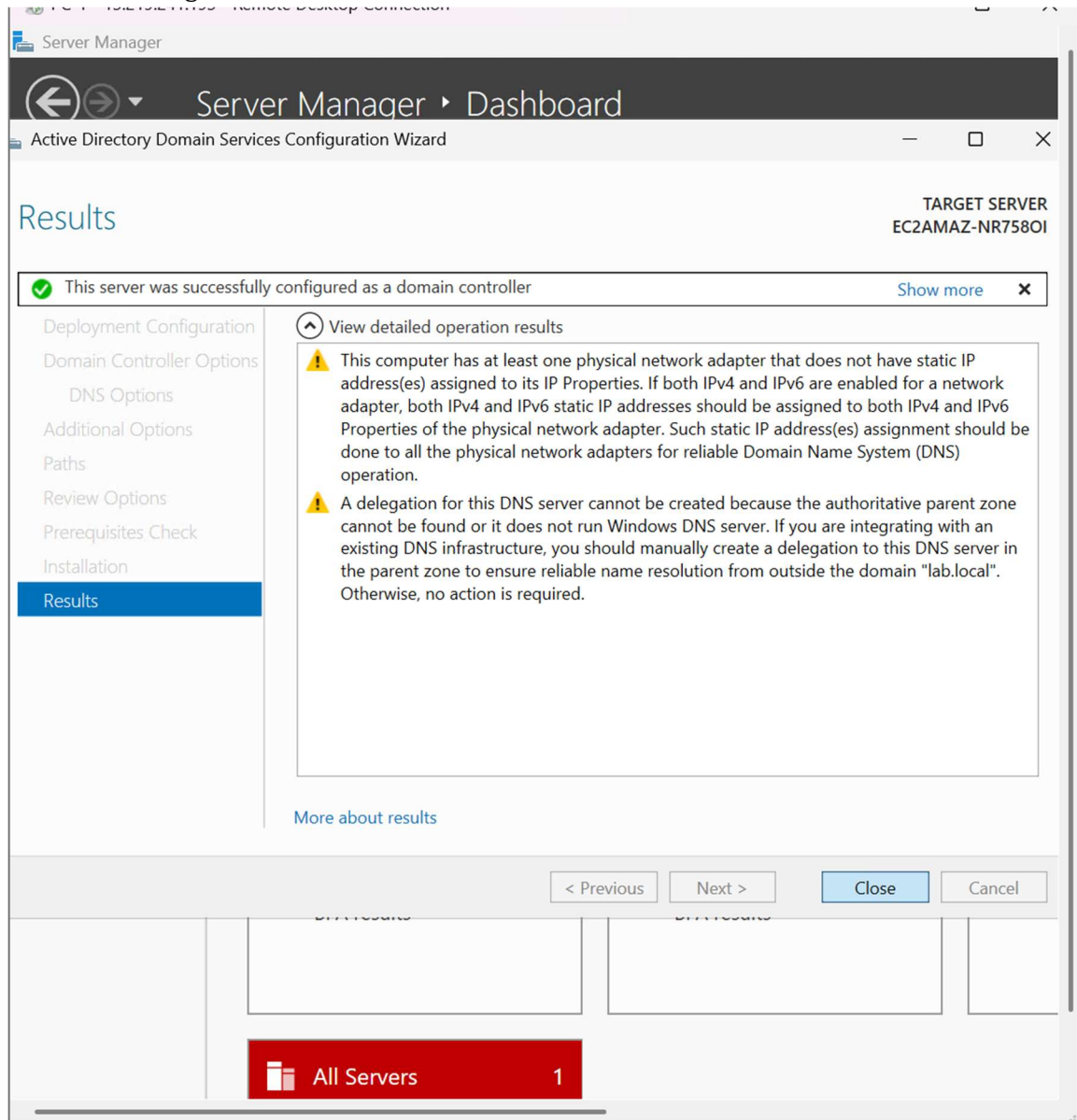
[More about Active Directory paths](#)

< Previous Next > Install Cancel

All Servers 1

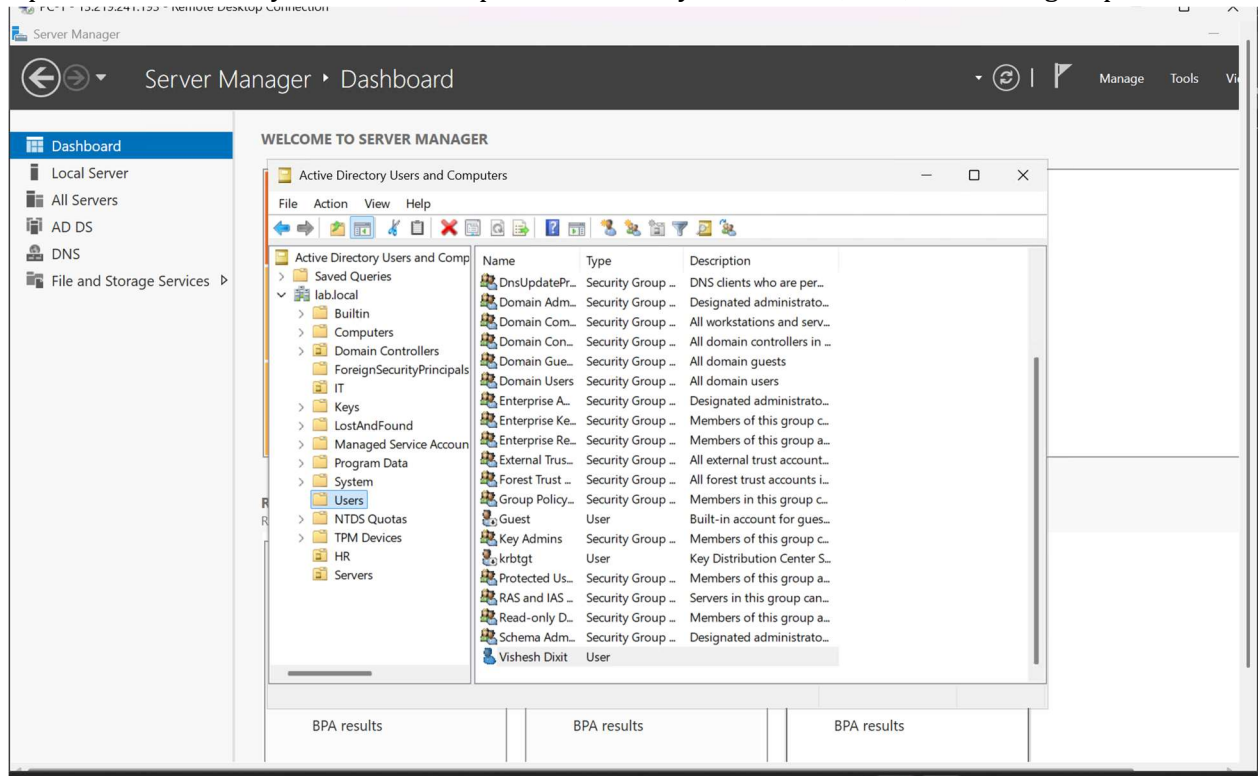
AD Step 6: Complete Installation

Review warnings, install AD DS and reboot.



AD Step 7: Verify Active Directory

Open Active Directory Users and Computers and verify the domain, OUs, users and groups.



Part 3: Join Additional EC2 Windows Instances

1. Launch another Windows Server or Windows 11 EC2 in the same VPC.
2. Place it in the same subnet or a routed subnet.
3. Configure Preferred DNS to the Domain Controller's private IP.
4. Verify connectivity using ping/nslookup.
5. Open System Properties > Change.
6. Select Domain and enter lab.local.
7. Authenticate using LAB\Administrator.
8. Restart the instance.
9. Verify the computer object appears in Active Directory Users and Computers.
10. Move the computer into the appropriate OU (Servers, IT, HR).

Skills Demonstrated

- AWS VPC
- Internet Gateway
- Route Tables
- Security Groups
- EC2
- Windows Server 2025
- Active Directory Domain Services
- DNS
- Domain Controller
- Domain Join
- Organizational Units
- User & Group Management